

Vishwateja Donikena

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EDUCATION

Vardhaman College of Engineering

Bachelor of Technology in Information Technology, JNTUH

Shamshabad, Hyderabad

2022-26, CGPA: 8.33

TECHNICAL SKILLS

Programming Languages : Python, SQL, C++.

AI/ML: LangChain, RAG, FAISS, HuggingFace, Groq API, Llama 3, Scikit-Learn, Pandas, NumPy

Full Stack : React, Node.js, Express.js, MongoDB, MySQL, HTML/CSS, REST APIs, FastAPI

Dev Tools: VS Code, Git, GitHub.

Relevant Coursework : Data Structures, Machine Learning, DBMS, Operating Systems.

RESEARCH AND PUBLICATIONS

RAG-Based Cardiac Decision Support System using LLMs

October 2025 - March 2026

- * Engineered a **Retrieval-Augmented Generation (RAG)** pipeline over a 128-page ESC cardiac guideline, enabling evidence-grounded clinical QA.
- * Benchmarked MiniLM vs. BioBERT embeddings achieved **85%** accuracy, **100%** recall, and **85%** F1-score, with BioBERT delivering superior domain-specific retrieval performance.
- * Built an end-to-end RAG system using Python, LangChain, FAISS, and LLaMA 3 (Groq API); implemented hybrid search and prompt engineering to deliver highly relevant, guideline-aligned responses.
- * Designed a structured evaluation pipeline with LLM-as-a-judge, ensuring consistent scoring of clinical correctness across diagnostic queries and improving the reliability of model evaluation.
- * Identified key failure modes (context under-retrieval and incomplete multi-step retrieval) and improved retrieval quality through chunking strategies and domain-specific embeddings.

PROJECTS

Safety-Aware Multi-Agent Financial Assistant | FastAPI AI Microservice

- * Built a **safety-aware multi-agent financial assistant** using **FastAPI**, implementing **4+ core workflows** including safety screening, intent classification, agent routing, and real-time Server-Sent Events (SSE) response streaming.
- * Implemented a rule-based **safety guard** covering **6+ harmful financial query categories**, achieving **95%+ harmful-query recall** on the evaluation dataset and preventing unsafe requests from reaching the LLM.
- * Developed an **intent classifier** with structured entity extraction and follow-up query handling, supporting routing across **8+ financial query types** with **85%+ routing accuracy** on labeled test cases.
- * Built a **Portfolio Health** agent to analyze holdings across **5 investor-profile datasets**, surfacing concentration risk, benchmark comparisons, empty-portfolio guidance, and beginner-friendly investment insights.
- * Added end-to-end automated test coverage for safety, routing, streaming, and portfolio analysis with **12/12 passing tests**, enabling reliable CI execution without external API credentials.

Real-Time Expert Session Booking System | MERN Web Application

- * Built a full-stack **expert session booking system** using **React, Node.js, Express.js, MongoDB Atlas, and Socket.io**, managing **50+ expert slots** and booking records.
- * Designed RESTful APIs for expert listing, booking creation, status updates, and email-based booking search, improving data flow and reducing manual coordination by **40%**.
- * Implemented **real-time slot synchronization** using Socket.io, instantly disabling booked slots across users and improving booking accuracy by **60%**.
- * Prevented duplicate bookings using a **MongoDB unique compound index** on expert, date, and time slot, ensuring race-condition-safe reservations.
- * Developed a responsive React frontend featuring search, category filters, pagination, form validation, and booking status tracking, improving usability and accessibility by **35%**.

CERTIFICATIONS

- IBM Full Stack Software Developer Professional Certificate
- Salesforce Certified Agentforce Specialist
- AWS Academy Cloud Foundations
- Data Science for Engineers (Elite)
- Data Visualization Simulation
- CCNAv7: Introduction to Networks

Coursera

Salesforce

AWS Academy

NPTEL

Tata Consultancy Services (TCS)

Cisco Networking Academy